STACKS ON SPACES

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1. Motivation and Stack Definition

In this course we often produced a set of isomorphism classes of things. For example, recall from Theorem 5.35 that given a topological group G and a space X, we have that $\check{H}^1(X, G)$ is in bijection with the set of isomorphism classes of principal G-bundles over X. The assignment of a space X to the set of isomorphism classes of principal G-bundles over X factors through the assignment of X to the groupoid of principal G-bundles over X, by postcomposing with taking isomorphism classes. Therefore, we should have a notion of a sheaf that outputs a groupoid.

Vaguely, we know that a sheaf of groupoids should be a functor $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ satisfying something like a sheaf condition, which is going to be a limit condition in Grpd. Based on our intuition from sheaves of abelian groups in this course, we might expect it to be the same as the sheaf condition.

Definition 1.1 (Sheaf condition)**.** Recall that the *sheaf condition* for a sheaf of abelian groups $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Ab}$ is that for any open subset $U \subseteq X$ and any open cover $\mathcal{U} = \{U_i\}$ of U, we have an isomorphism

$$F(U) \xrightarrow{\cong} \lim \left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \right).$$

Remark 1.2. It turns out that this is not the correct limit condition for a sheaf of groupoids, so we must define something new but similar called a stack which has the correct limit condition.

Definition 1.3 (Stack)**.** Let X be a space and $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor. We say F is a *stack* if for every open subset $U \subseteq X$ and every open cover $\mathcal{U} = \{U_i\}$ of U, the functor

$$F(U) \xrightarrow{\simeq} 2\text{-}\lim \left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

given by the universal property of the 2-limit in Grpd is an equivalence of categories.

Remark 1.4. We can also repackage this definition by introducing the category of descent data.

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Definition 1.5 (Descent data)**.** Let X be a space and $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor. For any open subset $U \subseteq X$ and any open cover $\mathcal{U} = \{U_i\}$ of U, we call the groupoid

$$\mathrm{Desc}(\mathcal{U}, F) = 2\text{-lim}\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

the category of *descent data*, which we will denote by $\mathrm{Desc}(\mathcal{U}, F)$. Then, F is a stack if $F(U)$ is equivalent to all categories of descent data, that is, if we have an equivalence of categories $F(U) \simeq \mathrm{Desc}(\mathcal{U}, F)$ for any open subset $U \subseteq X$ and any open cover $\mathcal{U} = \{U_i\}$ of U.

2. Grothendieck Topology and Sites

The definition of a stack in Definition 1.1 is for the small site associated to a space X. In this section, we will discuss the definition of a stack on an arbitrary site. Vaguely, a site is a category $\mathcal{C}$ equipped with something called a Grothendieck topology, which makes the objects of $\mathcal{C}$ behave like the open subsets of a topological space, so that we can define the notion of an "open cover".

Definition 2.1 (Grothendieck topology)**.** Let $\mathcal{C}$ be a category. A *Grothendieck topology* on $\mathcal{C}$ is the assignment to each object $U \in \mathcal{C}$ a collection of sets of morphisms $\{U_i \to U\}$, called *coverings of* U, so that the following conditions are satisfied:

(1) If $V \xrightarrow{\cong} U$ is an isomorphism, then the set $\{V \to U\}$ is a covering.
(2) If $\{U_i \to U\}$ is a covering and $V \to U$ is any morphism, then the fibered products $\{U_i \times_U V\}$ exist, and the collection of projections $\{U_i \times_U V \to V\}$ is a covering.
(3) If $\{U_i \to U\}$ is a covering, and for each index i we have a covering $\{V_{ij} \to U_i\}$ (here j varies on a set depending on i), the collection of composites $\{V_{ij} \to U_i \to U\}$ is a covering of U.

This definition is Definition 2.24 in Vistoli's *Notes on Grothendieck topologies, fibered categories and descent theory* [Vis04].

Definition 2.2 (Site)**.** A category equipped with a Grothendieck topology is called a *site*.

Definition 2.3 (Stack, arbitrary site)**.** Let $\mathcal{C}$ be a site and $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor. We say F is a *stack* if for every object $U \in \mathcal{C}$ and every covering $\{U_i \to U\}$, the functor

$$F(U) \xrightarrow{\simeq} 2\text{-lim}\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \times_U U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \times_U U_j \times_U U_k) \right)$$

given by the universal property of the 2-limit in Grpd is an equivalence of categories.

Example 2.4. The small site associated to a space X is the category $\mathrm{Open}(X)$ equipped with the Grothendieck topology associating each open subset $U \subseteq X$ to the collection of open covers of U. Note that if $U_i \hookrightarrow U$ and $U_j \hookrightarrow U$ are morphisms, the fibered product $U_i \times_U U_j = \{(x, y) \in U_i \times U_j \mid x = y\}$ is the intersection $U_i \cap U_j$. The definition of a stack on the small site associated to a space X is in Definition 1.1.

Example 2.5. The global classical topology is the site consisting of the category of topological spaces Top, equipped with the Grothendieck topology associating each space X to the collection of sets of jointly surjective open embeddings $\{U_i \hookrightarrow X\}$. A set of open embeddings being jointly surjective means the union of their images equals X, i.e. their images form an open cover of X. Note that by "open embedding" we mean an open continuous injective map $V \hookrightarrow X$. If we instead meant the inclusion of an open subspace $V \subseteq X$, this would not satisfy condition (1) of a Grothendieck topology. This is because if V is homeomorphic to X, then the homeomorphism $\{V \xrightarrow{\cong} X\}$ is not necessarily a covering since V is not necessarily a subspace of X. The definition of a stack on the global classical topology is the following.

Definition 2.6 (Stack, global classical topology)**.** Let Top be the global classical topology and $F : \mathrm{Top}^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor. We say F is a *stack* if for every space X and every set of jointly surjective open embeddings $\{U_i \hookrightarrow X\}$, the functor

$$F(X) \xrightarrow{\simeq} 2\text{-}\lim\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

given by the universal property of the 2-limit in Grpd is an equivalence of categories.

Remark 2.7. In what follows, we will mostly be concerned with stacks on the small site associated to a space, and occasionally on the global classical topology. Therefore, we will just write stack to mean a stack on the small site associated to a space, and if the site is different we will specify it.

3. 2-Categorical Notions

The definition of a stack we gave in Definition 1.1 involves contravariant pseudofunctors from the 1-category $\mathrm{Open}(X)$ to the (2,1)-category Grpd and 2-limits in Grpd. These are the 2-categorical notions where many of the interesting differences between stacks and sheaves arise. Because we do not want to assume any familiarity with 2-categories, we will introduce the requisite background and definitions.

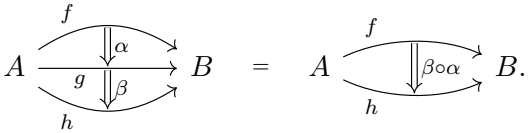
Definition 3.1 (2-category)**.** A *2-category* $\mathcal{C}$ is the data of:

(1) objects (or 0-cells) $A, B, C, \cdots \in \mathcal{C}$,
(2) 1-morphisms (or 1-cells) $f : A \to B$ between objects A, B, and
(3) 2-morphisms (or 2-cells) $\alpha : f \Rightarrow g$ between 1-morphisms $f, g : A \to B$.

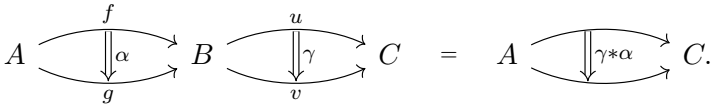
We require that:

- ▷ for every object A there exists an identity 1-morphism $\mathrm{id}_A : A \to A$, for every 1-morphism $f : A \to B$ there exists an identity 2-morphism $\mathrm{id}_f : f \Rightarrow f$,
- ▷ we can compose 1-morphisms and 2-morphisms, that is, for 1-morphisms $f : A \to B$, $g : B \to C$ there exists a 1-morphism $gf : A \to C$, and for 2-morphisms $\alpha : f \Rightarrow g$, $\beta : g \Rightarrow h$, where $f, g, h : A \to B$, there exists a 2-morphism $\beta \circ \alpha : f \Rightarrow h$,
- ▷ for every 1-morphism $f : A \to B$ we have $f\mathrm{id}_A = f = \mathrm{id}_B f$, and for every 2-morphism $\alpha : f \Rightarrow g$ we have $\alpha \circ \mathrm{id}_f = \alpha = \mathrm{id}_g \circ \alpha$,
- ▷ the compositions of 1-morphisms and 2-morphisms are associative.

In other words, the objects and 1-morphisms form a category called the *underlying category*, and for fixed objects A, B the 1-morphisms from $A \to B$ and the 2-morphisms between them form a category under the composition of 2-morphisms defined above, called *vertical composition*. The picture for vertical composition is:



There is another composition of 2-morphisms we must allow, called *horizontal composition*, defined as follows: for 2-morphisms $\alpha : f \Rightarrow g$, $\gamma : u \Rightarrow v$, where $f, g : A \to B$ and $u, v : B \to C$, there exists a 2-morphism $\gamma * \alpha : uf \Rightarrow vg$. We require that horizontal composition be associative, and that for every object A there exists an identity 2-morphism $\mathrm{id}_{\mathrm{id}_A} : \mathrm{id}_A \Rightarrow \mathrm{id}_A$, such that for any 2-morphism $\alpha : f \Rightarrow g$, where $f, g : A \to B$, we have $\mathrm{id}_{\mathrm{id}_B} * \alpha = \alpha = \alpha * \mathrm{id}_{\mathrm{id}_A}$. In other words, the objects and 2-morphisms form a category under horizontal composition. The picture for horizontal composition is:



We require that the two compositions satisfy the interchange law: for 2-morphisms $\alpha : f \Rightarrow g$, $\beta : g \Rightarrow h$ where $f, g, h : A \to B$, and $\gamma : u \Rightarrow v$, $\delta : v \Rightarrow w$ where $u, v, w : B \to C$, we have $(\delta * \beta) \circ (\gamma * \alpha) = (\delta \circ \gamma) * (\beta \circ \alpha)$. Finally, we require that for 1-morphisms $f : A \to B$ and $u : B \to C$, we have $\mathrm{id}_u * \mathrm{id}_f = \mathrm{id}_{uf}$.

This definition is adapted from 1.2 in Kelly and Street's *Review of the Elements of* 2*-Categories* [KS06].

Example 3.2. The standard example of a 2-category is the category of categories, denoted by Cat. The objects are categories $\mathcal{C}, \mathcal{D}, \cdots$, the 1-morphisms are functors $F : \mathcal{C} \to \mathcal{D}$, and the 2-morphisms are natural transformations $\alpha : F \Rightarrow G$.

Remark 3.3. The use of the term 0-cell to describe objects, 1-cell to describe 1-morphisms, and 2-cell to describe 2-morphisms, comes from the fact that when we consider a diagram in a 2-category it looks like a 2-dimensional cell-complex. For example, consider the following diagram:

$$A \underset{g}{\overset{f}{\Downarrow \alpha}} B,$$

in which we think of the 2-morphism α as "filling in" the space between the 1-morphisms f and g.

Definition 3.4 ((2,1)-category)**.** A *(2,1)-category* is a 2-category in which all 2-morphisms are 2-isomorphisms, that is, for every 2-morphism $\alpha : f \Rightarrow g$, there is an inverse 2-morphism $\beta : g \Rightarrow f$ such that $\beta \circ \alpha = \mathrm{id}_f$ and $\alpha \circ \beta = \mathrm{id}_g$.

Example 3.5. The standard example of a (2,1)-category is the category Grpd of groupoids. The objects are groupoids $\mathcal{C}, \mathcal{D}, \cdots$, the 1-morphisms are functors $F : \mathcal{C} \to \mathcal{D}$, and the 2-morphisms are natural isomorphisms $\alpha : F \cong G$.

Remark 3.6. Note that a 1-category is what we have been calling a category in this course.

Definition 3.7 (Pseudofunctor)**.** A contravariant *pseudofunctor* from a 1-category $\mathcal{C}$ to the 2-category of categories Cat, which we write as $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Cat}$, consists of the following data:

- ▷ For each object $C \in \mathcal{C}$ a category $F(C) \in \mathrm{Cat}$.
- ▷ For each morphism $f : A \to B$ in $\mathcal{C}$ a functor $F(f) : F(A) \to F(B)$ in Cat.
- ▷ For each object $C \in \mathcal{C}$ a specified natural isomorphism $\epsilon_C : F(\mathrm{id}_C) \cong \mathrm{id}_{F(C)}$, which we can picture in the following diagram:

$$F(C) \underset{\mathrm{id}_{F(C)}}{\overset{F(\mathrm{id}_C)}{\Downarrow \epsilon_C}} F(C).$$

- ▷ For each pair of composable morphisms $f : A \to B$, $g : B \to C$ in $\mathcal{C}$ a specified natural isomorphism $\alpha_{f,g} : F(gf) \cong F(f)F(g)$, which we can picture in the following diagram:

$$\begin{array}{ccccc} & & F(B) & & \\ & \swarrow^{F(f)} & \Downarrow \alpha_{f,g} & \nwarrow^{F(g)} & \\ F(A) & & \xleftarrow[F(gf)]{} & & F(C). \end{array}$$

These data are required to satisfy the following two coherence conditions:

(1) If $f : A \to B$ is a morphism in $\mathcal{C}$ and $\eta \in F(B)$ in an object in $F(B)$, we have

$$\alpha_{\mathrm{id}_A, f}(\eta) = \epsilon_A(F(f)(\eta)) : F(\mathrm{id}_A)F(f)(\eta) \xrightarrow{\cong} F(f)(\eta)$$

and

$$\alpha_{f, \mathrm{id}_B}(\eta) = F(f)\epsilon_B(\eta) : F(f)F(\mathrm{id}_B)(\eta) \xrightarrow{\cong} F(f)(\eta).$$

(2) For each triple of composable morphisms $f : A \to B$, $g : B \to C$, $h : C \to D$ in $\mathcal{C}$ and every object $\theta \in F(D)$, the following diagram in $F(A)$ commutes:

$$\begin{array}{ccc} F(f)(F)(g)F(h)(\theta) & \xrightarrow{\alpha_{f,g}(F(h)(\theta))} & F(gf)F(h)(\theta) \\ \Big\downarrow{\scriptstyle F(f)\alpha_{g.h}(\theta)} & & \Big\downarrow{\scriptstyle \alpha_{gf,h}(\theta)} \\ F(f)F(hg)(\theta) & \xrightarrow[\alpha_{f,hg}(\theta)]{} & F(hgf)(\theta). \end{array}$$

This definition is adapted from Definition 3.10 in Vistoli's *Notes on Grothendieck topologies, fibered categories and descent theory* [Vis04].

Remark 3.8. A contravariant pseudofunctor from a 1-category $\mathcal{C}$ to the (2,1)-category of groupoids Grpd, which we write as $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Grpd}$, is defined in the exact same way except that the category $F(C)$ for each object $C \in \mathcal{C}$ is a groupoid.

Definition 3.9 (2-limit)**.** We won't give an explicit definition of a *2-limit* in Grpd. For our purposes, it suffices to think of it as a limit whose cones commute up to coherent natural isomorphisms, and whose universal property is expressed by an equivalence of categories. In what follows, we'll just state the data of the 2-limits that we encounter.

Remark 3.10. The definition of a stack in Definition 1.1 is difficult to use for two main reasons:

(1) It's difficult to actually define a pseudofunctor from a 1-category to Grpd. In practice, it is not too difficult to define the assignment of objects to groupoids and morphisms to functors, but it is often tedious to specify the natural isomorphisms for composable pairs and to check that they satisfy the coherence conditions.
(2) It's difficult to work with 2-limits in Grpd because they require keeping track of the specified natural isomorphisms, which are often a lot of data, especially when compared to the data of our usual notion of a limit in a 1-category.

Example 3.11. Here's an example to see that the 2-limit of a diagram in Grpd consists of more data than the limit of the same diagram in a 1-category. Consider the diagram in the 1-category Set consisting of two maps $f : X \to Z$ and $g : Y \to Z$. The limit of this diagram is called the pullback, and it is the data of a set P and maps $p_1 : P \to X$, $p_2 : P \to Y$ such that

$$\begin{array}{ccc} P & \xrightarrow{p_1} & X \\ \Big\downarrow{\scriptstyle p_2} & \lrcorner & \Big\downarrow{\scriptstyle f} \\ Y & \xrightarrow[g]{} & Z \end{array}$$

commutes and such that P satisfies the universal property of a limit. Concretely, the set P is $P = X \times_Z Y = \{(x, y) \in X \times Y : f(x) = g(y)\}$ and the maps p_1, p_2 are projections onto the first and second component respectively.

Now, consider the diagram in Grpd consisting of two functors $F : \mathcal{A} \to \mathcal{C}$ and $G : \mathcal{B} \to \mathcal{C}$. The 2-limit of this diagram is called the 2-pullback, and it is the data of a groupoid $\mathcal{P}$ and functors $P_1 : \mathcal{P} \to \mathcal{A}$, $P_2 : \mathcal{P} \to \mathcal{B}$ such that

$$\begin{array}{ccc} \mathcal{P} & \xrightarrow{P_1} & \mathcal{A} \\ \Big\downarrow{\scriptstyle P_2} & \swarrow{\scriptstyle \alpha} & \Big\downarrow{\scriptstyle F} \\ \mathcal{B} & \xrightarrow[G]{} & \mathcal{C} \end{array}$$

commutes up to a specified natural isomorphism $\alpha : FP_1 \cong GP_2$, and such that $\mathcal{P}$ satisfies the universal property of a 2-limit. Concretely, the groupoid $\mathcal{P}$ is the data of:

- ▷ objects (A, B, φ), where $A \in \mathcal{A}$, $B \in \mathcal{B}$, and $\varphi : F(A) \xrightarrow{\cong} G(B)$ is an isomorphism in $\mathcal{C}$,
- ▷ isomorphisms $\psi : (A, B, \varphi) \xrightarrow{\cong} (A', B', \varphi')$ which are the data of isomorphisms $\psi_A : A \xrightarrow{\cong} A'$ in $\mathcal{A}$ and $\psi_B : B \xrightarrow{\cong} B'$ in $\mathcal{B}$ such that the following diagram in $\mathcal{C}$ commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{F(\psi_A)} & F(A') \\ \Big\downarrow{\scriptstyle \varphi} & & \Big\downarrow{\scriptstyle \varphi'} \\ G(B) & \xrightarrow[G(\psi_B)]{} & G(B'). \end{array}$$

The functor $P_1 : \mathcal{P} \to \mathcal{A}$ is the projection sending

$$(A, B, \varphi) \mapsto A \quad \text{and} \quad (\psi : (A, B, \varphi) \xrightarrow{\cong} (A', B', \varphi')) \mapsto (\psi_A : A \xrightarrow{\cong} A').$$

Similarly, the functor $P_2 : \mathcal{P} \to \mathcal{B}$ is the projection sending

$$(A, B, \varphi) \mapsto B \quad \text{and} \quad (\psi : (A, B, \varphi) \xrightarrow{\cong} (A', B', \varphi')) \mapsto (\psi_B : B \xrightarrow{\cong} B').$$

Therefore, for every object $(A, B, \varphi) \in \mathcal{P}$ we have an isomorphism $\varphi : FP_1(A, B, \varphi) = F(A) \xrightarrow{\cong} GP_2(A, B, \varphi) = G(B)$ in $\mathcal{C}$, such that for every isomorphism $\psi : (A, B, \varphi) \xrightarrow{\cong} (A', B', \varphi')$ in $\mathcal{P}$ the following diagram in $\mathcal{C}$ commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{\varphi} & G(B) \\ \Big\downarrow{\scriptstyle FP_1(\psi)=F(\psi_A)} & & \Big\downarrow{\scriptstyle G(\psi_B)=GP_2(\psi)} \\ F(A') & \xrightarrow[\varphi']{} & G(B'). \end{array}$$

Thus, we can see that the isomorphisms φ for all objects $(A, B, \varphi) \in \mathcal{P}$ are components of a natural isomorphism $FP_1 \cong GP_2$, which is the specified natural isomorphism $\alpha : FP_1 \cong GP_2$.

4. Triple Overlaps and the Cocycle Condition

We will eventually address the two difficulties with our definition of a stack by introducing an equivalent definition that's easier to work with. However, before doing this, we consider the following question that arises naturally when comparing the definition of a stack to that of a sheaf.

Question 4.1. Why do we need to consider triple overlaps as opposed to just double overlaps?

Answer 4.2. Let's consider a simple example that will help answer this question.

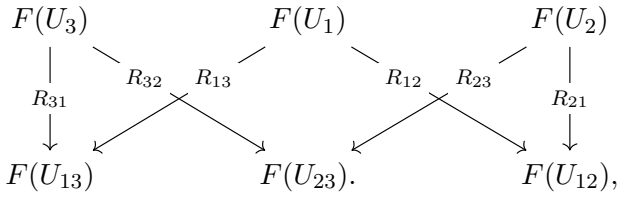
Example 4.3. Let X be a space, and suppose we have a three-object open cover $\mathcal{U} = \{U_1, U_2, U_3\}$. Let $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor. Then F is a stack if we have an equivalence of categories

$$F(X) \xrightarrow{\simeq} 2\text{-lim}\left(F(U_1) \times F(U_2) \times F(U_3) \rightrightarrows F(U_{12}) \times F(U_{13}) \times F(U_{23}) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; F(U_{123}) \right).$$

Note that we are using the shorthand notation $U_{ij} = U_i \cap U_j$ and $U_{ijk} = U_i \cap U_j \cap U_k$. Let's ignore the triple overlap U_{123} for now to see what happens. Therefore, we assume we have an equivalence of categories $F(X) \simeq E$, where

$$E = 2\text{-lim}\left(F(U_1) \times F(U_2) \times F(U_3) \rightrightarrows F(U_{12}) \times F(U_{13}) \times F(U_{23}) \right).$$

We can think of E as the 2-limit of the following diagram in Grpd:



where each functor $R_{ij} = F(U_{ij} \hookrightarrow U_i) : F(U_i) \to F(U_{ij})$ is induced by the inclusion $U_{ij} \hookrightarrow U_i$. Concretely, the groupoid E is the data of:

▷ objects which are 6-tuples of the form $(a_1, a_2, a_3, \phi_{12}, \phi_{13}, \phi_{23})$, where

$$a_1 \in F(U_1),\ a_2 \in F(U_2),\ a_3 \in F(U_3)$$

are local objects and

$$\phi_{12} : R_{12}(a_1) \xrightarrow{\cong} R_{21}(a_2) \text{ in } F(U_{12}),$$
$$\phi_{13} : R_{13}(a_1) \xrightarrow{\cong} R_{31}(a_1) \text{ in } F(U_{13}),$$
$$\phi_{23} : R_{23}(a_2) \xrightarrow{\cong} R_{32}(a_3) \text{ in } F(U_{23}),$$

are specified isomorphisms between local objects on the double overlaps,

▷ isomorphisms

$$f : (a_1, a_2, a_3, \phi_{12}, \phi_{13}, \phi_{23}) \xrightarrow{\cong} (a_1', a_2', a_3', \phi_{12}', \phi_{13}', \phi_{23}'),$$

which are the data of local isomorphisms

$$f_1 : a_1 \xrightarrow{\cong} a_1' \text{ in } F(U_1), \quad f_2 : a_2 \xrightarrow{\cong} a_2' \text{ in } F(U_2), \quad \text{and} \quad f_3 : a_3 \xrightarrow{\cong} a_3' \text{ in } F(U_3)$$

such that the following diagrams, respectively in $F(U_{12})$, $F(U_{13})$, and $F(U_{23})$, commute:

$$\begin{array}{ccc} R_{12}(a_1) & \xrightarrow[\cong]{R_{12}(f_1)} & R_{12}(a_1') \\ {\scriptstyle \phi_{12}}\downarrow{\scriptstyle \cong} & & {\scriptstyle \cong}\downarrow{\scriptstyle \phi_{12}'} \\ R_{21}(a_2) & \xrightarrow[R_{21}(f_2)]{\cong} & R_{21}(a_2') \end{array} \qquad \begin{array}{ccc} R_{13}(a_1) & \xrightarrow[\cong]{R_{13}(f_1)} & R_{13}(a_1') \\ {\scriptstyle \phi_{13}}\downarrow{\scriptstyle \cong} & & {\scriptstyle \cong}\downarrow{\scriptstyle \phi_{13}'} \\ R_{31}(a_3) & \xrightarrow[R_{31}(f_3)]{\cong} & R_{31}(a_3') \end{array}$$

$$\begin{array}{ccc} R_{23}(a_2) & \xrightarrow[\cong]{R_{23}(f_2)} & R_{23}(a_2') \\ {\scriptstyle \phi_{23}}\downarrow{\scriptstyle \cong} & & {\scriptstyle \cong}\downarrow{\scriptstyle \phi_{23}'} \\ R_{32}(a_3) & \xrightarrow[R_{32}(f_3)]{\cong} & R_{32}(a_3'). \end{array}$$

Let's see what happens to local objects when we restrict to the triple overlap $F(U_{123})$. Consider, for example, the pair of local objects $a_1 \in F(U_1)$ and $a_3 \in F(U_3)$. We claim that there are two isomorphisms between their respective restrictions to the triple overlap. Note that in what follows, the functors $T_i = F(U_{123} \hookrightarrow U_{jk}) : F(U_{jk}) \to F(U_{123})$ are induced by the inclusions $U_{jk} \hookrightarrow U_{123}$.

(1) If we start by restricting both to the double overlap $F(U_{13})$, we have the specified isomorphism $\phi_{13} : R_{13}(a_1) \xrightarrow{\cong} R_{31}(a_3)$. Then, if we restrict to the triple overlap $F(U_{123})$ by applying the functor $T_2 : F(13) \to F(U_{123})$, we obtain an isomorphism

$$T_2(\phi_{13}) : T_2(R_{13}(a_1)) \xrightarrow{\cong} T_2(R_{31}(a_3)) \quad \text{in } F(U_{123}).$$

(2) If we start by restricting a_3 to $F(U_{23})$ and a_1 to $F(U_{12})$, we have $\phi_{23} : R_{23}(a_2) \xrightarrow{\cong} R_{32}(a_3)$ and $\phi_{12} : R_{12}(a_1) \xrightarrow{\cong} R_{21}(a_2)$. Then, if we restrict to $F(U_{123})$ by applying $T_1 : F(U_{23}) \to F(U_{123})$ and $T_3 : F(U_{12}) \to F(U_{123})$, we have isomorphisms $T_1(\phi_{23}) : T_1(R_{23}(a_2)) \xrightarrow{\cong} T_1(R_{32}(a_3))$ and $T_3(\phi_{12}) : T_3(R_{12}(a_1)) \xrightarrow{\cong} T_3(R_{21}(a_2))$ in $F(U_{123})$. Since $T_1(R_{23}(a_2)) \cong T_3(R_{21}(a_2))$ are canonically isomorphic by the pseudofunctoriality of F (which we will make precise later), we can compose these isomorphisms to obtain

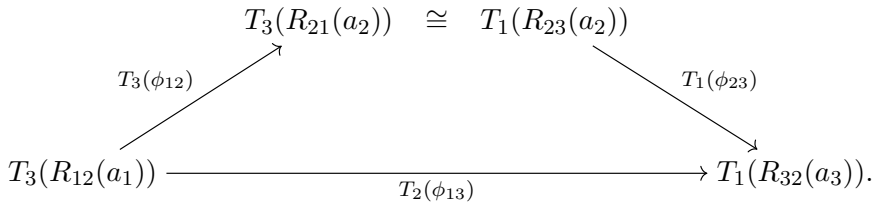
$$T_1(\phi_{23}) \circ T_3(\phi_{12}) : T_3(R_{12}(a_1)) \xrightarrow{\cong} T_1(R_{32}(a_3)) \quad \text{in } F(U_{123}),$$

where we are suppressing the canonical isomorphism between $T_1(R_{23}(a_2)) \cong T_3(R_{21}(a_2))$ in the notation.

(3) We can also start by restricting a_1 to $F(U_{12})$ and a_3 to $F(U_{13})$, and then restricting to $F(U_{123})$ by applying T_3 and T_2, so we are considering objects $T_3(R_{12}(a_1))$ and $T_2(R_{31}(a_3))$. We already have an isomorphism $T_1(\phi_{23}) \circ T_3(\phi_{12}) : T_3(R_{12}(a_1)) \xrightarrow{\cong} T_1(R_{32}(a_3))$, and $T_2(R_{31}(a_3)) \cong T_1(R_{32}(a_3))$ are canonically isomorphic since F is a pseudofunctor, so up to composing with this canonical isomorphism we have the same isomorphism as in (2).

(4) Similarly, we can start by restricting a_1 to $F(U_{13})$ and a_3 to $F(U_{23})$, and then restricting to $F(U_{123})$ by applying T_2 and T_1, so we are considering objects $T_2(R_{13}(a_1))$ and $T_1(R_{32}(a_3))$. We already have an isomorphism $T_1(\phi_{23}) \circ T_3(\phi_{12}) : T_3(R_{12}(a_1)) \xrightarrow{\cong} T_1(R_{32}(a_3))$, and $T_2(R_{13}(a_1)) \cong T_3(R_{12}(a_1))$ are canonically isomorphic since F is a pseudofunctor, so up to composing with this canonical isomorphism we have the same isomorphism as in (2).

Thus, we have shown that there are two isomorphisms between a_1 and a_3 restricted to the triple overlap (up to composing with canonical isomorphisms given by the pseudofunctoriality of F). However, such an isomorphism should be uniquely specified, so we want these two isomorphisms to agree, that is, we want the following diagram in $F(U_{123})$ to commute:



We write this concisely as a "cocycle condition" on the triple overlap in the following way:

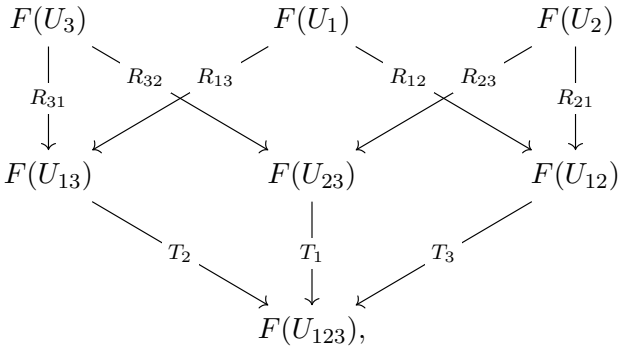
$$T_1(\phi_{23}) \circ T_3(\phi_{12}) = T_2(\phi_{13}),$$

where we are suppressing the canonical isomorphism $T_1(R_{23}(a_2)) \cong T_3(R_{21}(a_2))$ in the notation. Unfortunately, there's no reason this diagram needs to commute based on the data we have, so we are missing something. What we are missing is precisely the triple overlap in the 2-limit condition.

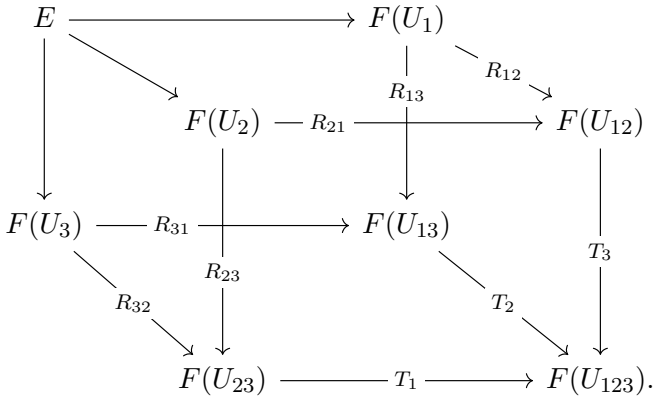
Suppose now that we have an equivalence of categories $F(X) \simeq E$, where

$$E = \text{2-lim}\left(F(U_1) \times F(U_2) \times F(U_3) \rightrightarrows \prod F(U_{12}) \times F(U_{13}) \times F(U_{23}) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod F(U_{123}) \right).$$

We can think of E as the 2-limit of the following diagram in Grpd:



so E fits into the following cube diagram in Grpd that commutes up to specified natural isomorphisms:



Concretely, the groupoid E is the data of:

(1) objects which are 6-tuples of the form $(a_1, a_2, a_3, \phi_{12}, \phi_{13}, \phi_{23})$, where

$$a_1 \in F(U_1),\ a_2 \in F(U_2),\ a_3 \in F(U_3)$$

are local objects, and

$$\phi_{12} : R_{12}(a_1) \xrightarrow{\cong} R_{21}(a_2) \text{ in } F(U_{12}),$$
$$\phi_{13} : R_{13}(a_1) \xrightarrow{\cong} R_{31}(a_3) \text{ in } F(U_{13}),$$
$$\phi_{23} : R_{23}(a_2) \xrightarrow{\cong} R_{32}(a_3) \text{ in } F(U_{23}),$$

are specified isomorphisms between local objects on the double overlaps which satisfy the cocycle condition on the triple overlap:

$$T_1(\phi_{23}) \circ T_3(\phi_{12}) = T_2(\phi_{13}),$$

(2) isomorphisms which are defined in the same way as above.

Therefore, we can see that including the triple overlap $F(U_{123})$ in the diagram does not add any additional data to the 2-limit E. What it does add is the condition that the isomorphisms on double overlaps $\phi_{12}, \phi_{13}, \phi_{23}$ satisfy the cocycle condition on the triple overlap, which is exactly what we were missing.

Let's try to give some intuition for why this is the case. Recall that E fits into the cube diagram above, which commutes up to specified natural isomorphisms. Let $(a_1, a_2, a_3, \phi_{12}, \phi_{13}, \phi_{23}) \in E$ be an arbitrary object, and let's consider the different ways this object can travel through the cube to $F(U_{123})$. First, note that the functors $E \to F(U_1)$, $E \to F(U_2)$, and $E \to F(U_3)$ send

$(a_1, a_2, a_3, \phi_{12}, \phi_{13}, \phi_{23})$ to the local objects a_1, a_2, and a_3. Now consider the pair of local objects $a_1 \in F(U_1)$ and $a_3 \in F(U_3)$. They each have two ways to travel through the diagram to $F(U_{123})$:

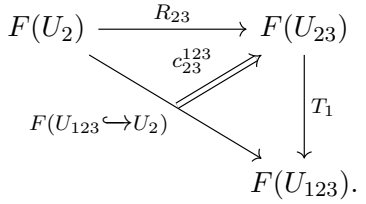
- ▷ for a_1, we write them as $T_3(R_{12}(a_1))$ and $T_2(R_{13}(a_1))$, and
- ▷ for a_3, we write them as $T_1(R_{32}(a_3))$ and $T_2(R_{31}(a_3))$.

We already examined all four possible isomorphisms between these objects, and we showed that there are only two (up to composing with some canonical isomorphisms). Since the cube diagram commutes up to specified natural isomorphisms, we know that there must be a unique isomorphism between these different ways to travel through the diagram, and we expressed this the cocycle condition $T_1(\phi_{23}) \circ T_3(\phi_{12}) = T_2(\phi_{13})$, which is now a condition on the data of the 2-limit E. Therefore, requiring that every way to travel through the cube be naturally isomorphic in a unique way amounts to imposing cocycle conditions. Following the same reasoning for the other possible pairs of local objects (a_1, a_2) and (a_2, a_3), we obtain two more cocycle conditions:

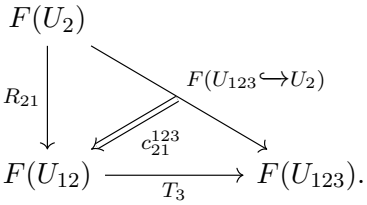
$$T_1(\phi_{23}^{-1}) \circ T_2(\phi_{13}) = T_3(\phi_{12}) \quad \text{and} \quad T_2(\phi_{13}) \circ T_3(\phi_{12}^{-1}) = T_1(\phi_{23}),$$

but they are redundant with the first one, so we omit them in the definition of the 2-limit E.

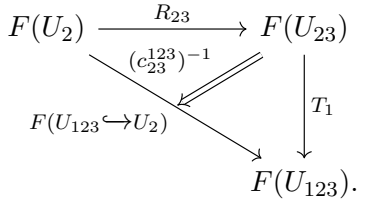
Remark 4.4. As mentioned, when writing the cocycle condition in the above example, we suppressed canonical isomorphisms given by the pseudofunctoriality of F, essentially treating them as a strict equalities. Let's be more precise about what these canonical isomorphisms are, by considering the first one that appeared in the example, between $T_1(R_{23}(a_2)) \cong T_3(R_{21}(a_2))$. Consider the composable pair of inclusions $U_{123} \hookrightarrow U_{23} \hookrightarrow U_2$. Since F is a contravariant pseudofunctor, we have a specified natural isomorphism $c_{23}^{123} : F(U_{123} \hookrightarrow U_{23} \hookrightarrow U_2) \cong F(U_{123} \hookrightarrow U_{23}) \circ F(U_{23} \hookrightarrow U_2) = T_1 \circ R_{23}$, which we can picture in:



Similarly, for the composable pair of inclusions $U_{123} \hookrightarrow U_{12} \hookrightarrow U_2$, we have a specified natural isomorphism $c_{21}^{123} : F(U_{123} \hookrightarrow U_{12} \hookrightarrow U_2) \cong F(U_{123} \hookrightarrow U_{12}) \circ F(U_{12} \hookrightarrow U_2) = T_3 \circ R_{21}$, which we can picture in:



Then, c_{23}^{123} has an inverse $(c_{23}^{123})^{-1} : T_1 \circ R_{23} \cong F(U_{123} \hookrightarrow U_{23} \hookrightarrow U_2)$ which we can picture in:



Since $U_{123} \hookrightarrow U_{23} \hookrightarrow U_2 = U_{123} \hookrightarrow U_{12} \hookrightarrow U_2 = U_{123} \hookrightarrow U_2$, we have $F(U_{123} \hookrightarrow U_{23} \hookrightarrow U_2) = F(U_{123} \hookrightarrow U_{12} \hookrightarrow U_2)$, so we can horizontally compose $(c_{23}^{123})^{-1}$ with c_{21}^{123} to obtain a natural

isomorphism $c_{21}^{123} \circ (c_{23}^{123})^{-1} : T_1 \circ R_{23} \cong T_3 \circ R_{21}$ which we can picture in the following square:

$$\begin{array}{ccc} F(U_2) & \xrightarrow{R_{23}} & F(U_{23}) \\ {\scriptstyle R_{21}}\downarrow & \swarrow^{c_{21}^{123}\circ(c_{23}^{123})^{-1}} & \downarrow{\scriptstyle T_1} \\ F(U_{12}) & \xrightarrow[T_3]{} & F(U_{123}). \end{array}$$

The component of $c_{21}^{123} \circ (c_{23}^{123})^{-1}$ at $a_2 \in F(U_2)$ is the canonical isomorphism $(c_{21}^{123} \circ (c_{23}^{123})^{-1})(a_2) : T_1(R_{23}(a_2)) \xrightarrow{\cong} T_3(R_{21}(a_2))$ that we suppressed in the cocycle condition. Strictly speaking then, the first cocycle condition should have be written as:

$$T_1(\phi_{23}) \circ (c_{21}^{123} \circ (c_{23}^{123})^{-1})(a_2) \circ T_3(\phi_{12}) = T_2(\phi_{13}),$$

such that the following diagram in $F(U_{123})$ commutes:

$$\begin{array}{ccc} T_3(R_{21}(a_2)) & \xrightarrow{(c_{21}^{123}\circ(c_{23}^{123})^{-1})(a_2)} & T_1(R_{23}(a_2)) \\ {\scriptstyle T_3(\phi_{12})}\uparrow & & \downarrow{\scriptstyle T_1(\phi_{23})} \\ T_3(R_{12}(a_1)) & \xrightarrow[T_2(\phi_{13})]{} & T_1(R_{32}(a_3)). \end{array}$$

Remark 4.5. In what follows, there may be other conditions that hold up to composing with canonical isomorphisms given by the pseudofunctoriality of F, but which are written such that they are suppressed in the notation, essentially treating them as strict equalities. We will try to be careful to point out when this is the case, as we did in the above example.

Remark 4.6. The above example also shows the large amount of data we need to keep track of when computing a 2-limit in Grpd.

Notation 4.7. We want to introduce some more convenient notation to write restrictions of objects. Let X be a space and $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ a pseudofunctor. Suppose $V \hookrightarrow U$ in $\mathrm{Open}(X)$, which induces a restriction functor $F(V \hookrightarrow U) : F(U) \to F(V)$. For each object $a \in F(U)$ and each morphism $f : a \to b$ in $F(U)$, we write their respective restrictions to $F(V)$ as $F(V \hookrightarrow U)(a) = a\mid_V$ and $F(V \hookrightarrow U)(f) = f\mid_V : a\mid_V \to b\mid_V$. We also continue using the shorthand notation $U_{ij} = U_i \cap U_j$ and $U_{ijk} = U_i \cap U_j \cap U_k$.

Remark 4.8. For reference, we will state the data of an arbitrary category of descent data.

Definition 4.9 (Descent data)**.** For a space X, a pseudofunctor $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$, an open subset $U \subset X$, and an open cover $\mathcal{U} = \{U_i\}$ of U, the category of *descent data*

$$\mathrm{Desc}(\mathcal{U}, F) = \text{2-lim}\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

is the groupoid with the data of:

(1) objects which are tuples of the form (a_i, ϕ_{ij}), where $a_i \in F(U_i)$ are local objects and $\phi_{ij} : a_i\mid_{U_{ij}} \xrightarrow{\cong} a_j\mid_{U_{ij}}$ in $F(U_{ij})$ are specified isomorphisms between local objects on the double overlaps, with inverses satisfying $\phi_{ij}^{-1} = \phi_{ji}$, and which satisfy the following cocycle condition on triple overlaps:

$$\phi_{jk}\mid_{U_{ijk}} \circ \phi_{ij}\mid_{U_{ijk}} = \phi_{ik}\mid_{U_{ijk}} \text{ in } F(U_{ijk}),$$

(up to composing with canonical isomorphisms given by pseudofunctoriality of F),

(2) isomorphisms $f : (a_i, \phi_{ij}) \xrightarrow{\cong} (a_i', \phi_{ij}')$, which are the data of local isomorphisms $f_i : a_i \xrightarrow{\cong} a_i'$ in $F(U_i)$ such that the following diagram in the each groupoid $F(U_{ij})$ commutes:

$$\begin{array}{ccc} a_i\,|_{U_{ij}} & \xrightarrow[\cong]{f_i|_{U_{ij}}} & a_i'\,|_{U_{ij}} \\ \Big\downarrow{\scriptstyle \phi_{ij}\,\cong} & & \Big\downarrow{\scriptstyle \cong\,\phi_{ij}'} \\ a_j\,|_{U_{ij}} & \xrightarrow[f_j|_{U_{ij}}]{\cong} & a_j'\,|_{U_{ij}}\ . \end{array}$$

5. Stacks from Prestacks

Our goal is to eventually give concrete examples of stacks and prove they are in fact stacks. However, to do this using the definition we gave in Definition 1.1, we would need to define the pseudofunctor $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$, compute the 2-limit $\mathrm{Desc}(\mathcal{U}, F)$ for any category of descent data, and show there is an equivalence of categories between $F(U)$ and $\mathrm{Desc}(\mathcal{U}, F)$. We discussed in Remark 3.10 why this is difficult to do. In particular, it is tedious to show that the pseudofunctor we want to define satisfies the two coherence conditions. Therefore, we will introduce an equivalent definition of a stack, which in some cases will be easier to work with. We start by defining something called a prestack, and then we will define a "stack condition" for a prestack to be a stack.

Definition 5.1 (Prestack)**.** Let X be a space. A *prestack* over $\mathrm{Open}(X)$ is a 1-category $\mathcal{F}$ and a functor $\pi : \mathcal{F} \to \mathrm{Open}(X)$ satisfying the following two conditions:

(1) Pullbacks exist: if $U \hookrightarrow V$ in $\mathrm{Open}(X)$ and $y \in \pi^{-1}(V)$, then there exists $x \in \pi^{-1}(U)$ and $f : x \to y$ so that $\pi(f) : U \hookrightarrow V$. We say $f : x \to y$ is a morphism over $U \hookrightarrow V$, and we have

$$\begin{array}{ccc} x & \overset{f}{\dashrightarrow} & y \\ {\scriptstyle\pi}\,\vdots & & \Big\downarrow{\scriptstyle\pi} \\ U & \hookrightarrow & V. \end{array}$$

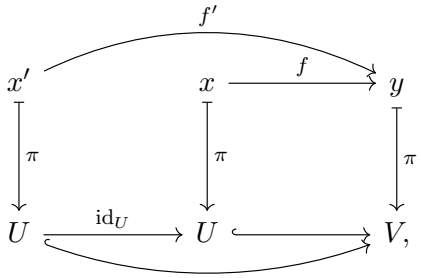
(2) Universal property for pullbacks: suppose $U \hookrightarrow V \hookrightarrow W$ in $\mathrm{Open}(X)$ and $z \in \pi^{-1}(W)$, then we know by (1) that there exists $x \in \pi^{-1}(U)$, $y \in \pi^{-1}(V)$, $f : x \to z$, and $g : y \to z$ so that $\pi(f) : U \hookrightarrow W$ and $\pi(g) : V \hookrightarrow W$. The universal property for pullbacks says that there exists a unique $h : x \to y$ such that $\pi(h) : U \hookrightarrow V$ and that the following diagram commutes:

$$\begin{array}{ccccc} & & \overset{f}{\frown} & & \\ x & \overset{h}{\dashrightarrow} & y & \xrightarrow{\quad g\quad} & z \\ \Big\downarrow{\scriptstyle\pi} & & \Big\downarrow{\scriptstyle\pi} & & \Big\downarrow{\scriptstyle\pi} \\ U & \hookrightarrow & V & \hookrightarrow & W. \end{array}$$

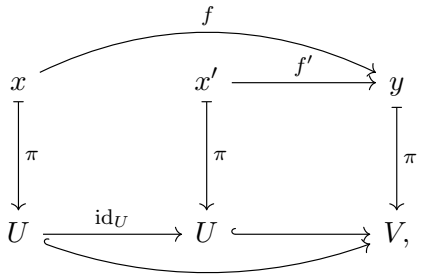
This definition is adapted from Definition 1 in Alper's *Evolution of Stacks and Moduli* [Alp].

Proposition 5.2. The universal property for pullbacks (2) implies that the pullbacks given by (1) are unique up to isomorphism.

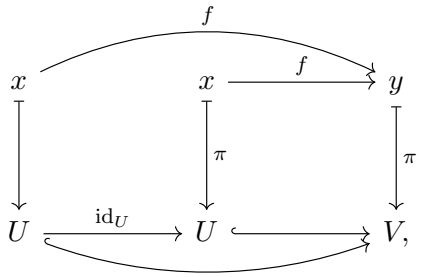
Proof. Let $U \hookrightarrow V$ in $\mathrm{Open}(X)$ and $y \in \pi^{-1}(V)$, then there exists $x \in \pi^{-1}(U)$ and $f : x \to y$ so that $\pi(f) : U \hookrightarrow V$. We want to show that x is unique up to isomorphism. Suppose that we also have $x' \in \pi^{-1}(U)$ and $f' : x' \to y$ so that $\pi(f') : U \hookrightarrow V$. Therefore, we have the following diagram:



so by the universal property for pullbacks there exists a unique $g : x' \to x$ such that $\pi(g) : U \xrightarrow{\mathrm{id}_U} U$ and that the above diagram including g commutes, so we can see that $f \circ g = f'$. By swapping x and x', we also have the following diagram:



so by the universal property for pullbacks there exists a unique $h : x \to x'$ such that $\pi(h) : U \xrightarrow{\mathrm{id}_U} U$ and that the above diagram including h commutes, so we can see that $f' \circ h = f$. Then, if we consider the following diagram:



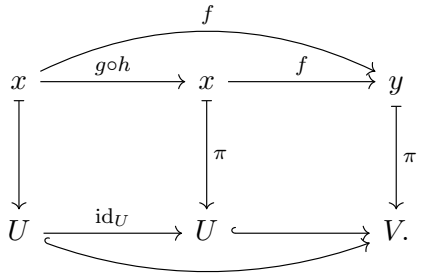
by the universal property for pullbacks there exists a unique $i : x \to x$ such that $\pi(i) : U \xrightarrow{\mathrm{id}_U} U$ and that the above diagram including i commutes. We can clearly see that the identity morphism $\mathrm{id}_x : x \to x$ satisfies these conditions, and so it is the unique morphism $i = \mathrm{id}_x$. However, we also have that the morphism $g \circ h : x \to x$ satisfies

$$\pi(g \circ h) = \pi(g) \circ \pi(h) = \mathrm{id}_U \circ \mathrm{id}_U = \mathrm{id}_U$$

and

$$f \circ (g \circ h) = (f \circ g) \circ h = f' \circ h = f,$$

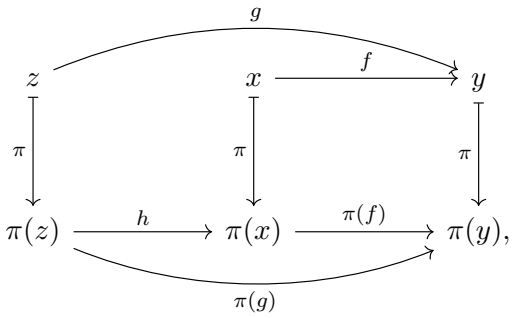
such that the following diagram commutes:



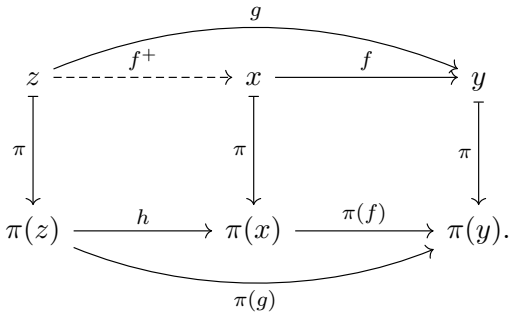
Therefore, we have shown that $g \circ h$ also satisfies the conditions of the universal property for pullbacks, and so by uniqueness we have $i = g \circ h = \mathrm{id}_x$. Following the same reasoning, we can show that $h \circ g = \mathrm{id}_{x'}$ as well. Thus, we have shown that $g : x' \xrightarrow{\cong} x$ is an isomorphism, and so the pullback x is unique up to isomorphism. □

Remark 5.3. We will now discuss Grothendieck fibrations, with the goal of showing that prestacks over $\mathrm{Open}(X)$ are the same as fibrations over $\mathrm{Open}(X)$. Since we eventually want to use a theorem from Johnson and Yau's *2-Dimensional Categories* [JY21] relating pseudofunctors from $\mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Cat}$ to fibrations over $\mathrm{Open}(X)$, we will adapt their relevant definitions. We start with two definitions adapted from Definition 9.1.1 in [JY21].

Definition 5.4 (Cartesian morphism)**.** Let $\pi : \mathcal{F} \to \mathrm{Open}(X)$ be a functor between 1-categories. A morphism $f : x \to y$ in $\mathcal{F}$ is called a *Cartesian morphism* if for every $g : z \to y$ in $\mathcal{F}$ and $h : \pi(z) \to \pi(x)$ such that $\pi(g) = h\pi(f)$, i.e. such that the bottom part of this diagram commutes:



there exists a unique $f^+ : z \to x$ in $\mathcal{F}$ such that $\pi(f^+) = f$ and $g = f^+ f$, i.e. such that the top part of this diagram commutes:



In the language of [JY21], this says that for a morphism $f : x \to y$ in $\mathcal{F}$, every pre-raise $\langle f, g; h\rangle$ has a unique raise.

Remark 5.5. This is the universal property for pullbacks in our definition of a prestack.

Definition 5.6 (Grothendieck fibration)**.** A functor $\pi : \mathcal{F} \to \mathrm{Open}(X)$ between 1-categories is called a *Grothendieck fibration* (or just a *fibration*) if for every $y \in \mathcal{F}$ and $U \hookrightarrow V$ in $\mathrm{Open}(X)$ with $\pi(y) = V$, there exists $x \in \mathcal{F}$ and a Cartesian morphism $f : x \to y$ in F called the *lift* such that $\pi(x) = U$ and $\pi(f) : U \hookrightarrow V$, i.e. such that the following diagram commutes:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ U & \hookrightarrow & V. \end{array}$$

In the language of the [JY21], this says that every pre-lift $\langle y; U \hookrightarrow V \rangle$ has a Cartesian lift, i.e. a lift which is a Cartesian morphism.

Remark 5.7. The existence of a lift is the fact that pullbacks exist in our definition of a prestack. The fact that the lift is Cartesian is the universal property for pullbacks. Therefore, we can see that a functor $\pi : \mathcal{F} \to \mathrm{Open}(X)$ between 1-categories is a prestack if and only if it is a fibration, so prestacks over $\mathrm{Open}(X)$ are exactly the same thing as fibrations over $\mathrm{Open}(X)$, just with different names.

Definition 5.8 (Category of fibrations)**.** There is a *category of fibrations* (or equivalently prestacks) over $\mathrm{Open}(X)$, which we will denote by $\mathrm{Fib}(\mathrm{Open}(X)) = \mathrm{Prestack}(\mathrm{Open}(X))$, which is the data of:

- ▷ objects which are fibrations (prestacks) $p : \mathcal{F} \to \mathrm{Open}(X)$, $q : \mathcal{G} \to \mathrm{Open}(X)$, $\cdots$, and
- ▷ morphisms $p \to q$ which are functors $F : \mathcal{F} \to \mathcal{G}$ that commute with the projections, that is, such that the following diagrams commute:

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{F} & \mathcal{G} \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & \mathrm{Open}(X), & \end{array}$$

and which send Cartesian morphisms in $\mathcal{F}$ to Cartesian morphisms in $\mathcal{G}$. These are called *Cartesian functors*. This definition is adapted from Definition 9.1.14 and Theorem 9.1.18 in [JY21].

Remark 5.9. The reason we are interested in prestacks is that there is an equivalence of categories between prestacks over $\mathrm{Open}(X)$ and pseudofunctors from $\mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Cat}$. Therefore, instead of considering a pseudofunctor $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$, which is difficult to work with as discussed in Remark 3.10, we can always just consider its corresponding prestack over $\mathrm{Open}(X)$ under this equivalence, which we state in the following theorem.

Theorem 5.10. There is an equivalence of categories

$$\mathrm{Fun}^{\mathrm{ps}}(\mathrm{Open}(X)^{\mathrm{op}}, \mathrm{Cat}) \simeq \mathrm{Prestack}(\mathrm{Open}(X))$$

given by the Grothendieck Construction.

Proof. We will use the fact that the Grothendieck construction is an equivalence of categories

$$\mathrm{Fun}^{\mathrm{ps}}(\mathrm{Open}(X)^{\mathrm{op}}, \mathrm{Cat}) \underset{\simeq}{\xrightarrow{\int}} \mathrm{Fib}(\mathrm{Open}(X))$$

from Theorem 10.6.16 in [JY21]. Then, since $\mathrm{Prestack}(\mathrm{Open}(X)) = \mathrm{Fib}(\mathrm{Open}(X))$, we have

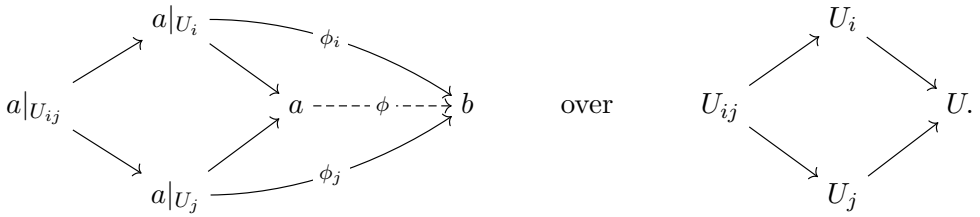
$$\mathrm{Fun}(\mathrm{Open}(X)^{\mathrm{op}}, \mathrm{Cat}) \simeq \mathrm{Prestack}(\mathrm{Open}(X)) = \mathrm{Fib}(\mathrm{Open}(X)).$$

□

Remark 5.11. We can now define stacks as prestacks satisfying two "stack conditions".

Definition 5.12 (Stack, prestack version)**.** We say a prestack $\pi : \mathcal{F} \to \mathrm{Open}(X)$ is a *stack* if:

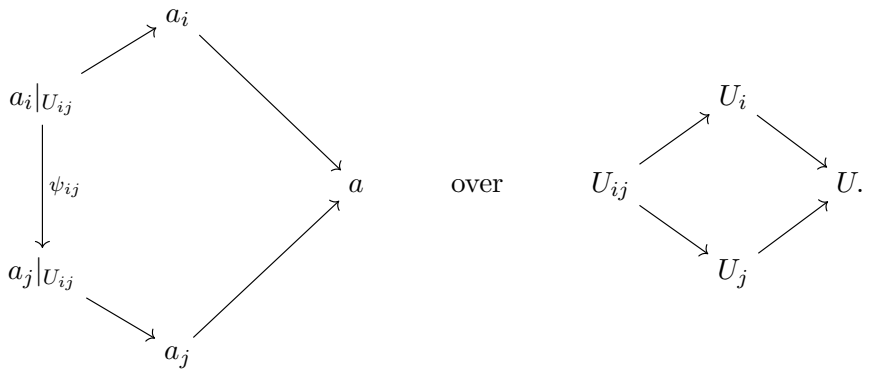
(1) Morphisms glue: for $a, b \in \mathcal{F}$ so that $\pi(a) = \pi(b) = U$, an open cover $\mathcal{U} = \{U_i\}$ of U, and morphisms $\phi_i : a|_{U_i} \to b\,|_{U_i}$ such that $\phi_i|_{U_{ij}} = \phi_j|_{U_{ij}}$, there exists a unique morphism $\phi : a \to b$ so that $\pi(\phi) = \mathrm{id}_U$ and $\phi\,|_{U_i} = \phi_i$. We can picture this as



(2) Objects glue: for $a_i \in \pi^{-1}(U_i)$ and isomorphisms $\psi_{ij} : a_i|_{U_{ij}} \xrightarrow{\cong} a_j|_{U_{ij}}$ satisfying the cocycle condition on triple overlaps:

$$\psi_{jk}|_{U_{ijk}} \circ \psi_{ij}|_{U_{ijk}} = \psi_{ik}|_{U_{ijk}},$$

there exists an object $a \in \pi^{-1}(U)$ and isomorphisms $\phi_i : a|_{U_i} \xrightarrow{\cong} a_i$ so that $\pi(\phi_i) = \mathrm{id}_{U_i}$ and $\phi_j|_{U_{ij}} = \psi_{ij} \circ \phi_i|_{U_{ij}}$. We can picture this as



This definition is adapted from Definition 3 in Alper's *Evolution of Stacks and Moduli* [Alp].

Question 5.13. Now that we've defined stacks over $\mathrm{Open}(X)$ as prestacks for which objects and morphisms glue, we should ask which pseudofunctors from $\mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Cat}$ they correspond to under the equivalence of categories in Theorem 5.10.

Answer 5.14. They should correspond to pseudofunctors $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ satisfying the 2-limit condition in our first definition of a stack, which is in Definition 1.1, so that our two definitions of a stack are equivalent. We state and prove this in the following theorem.

Theorem 5.15. Let $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor corresponding to a prestack $\pi : \mathcal{F} \to \mathrm{Open}(X)$ under the equivalence of categories in Theorem 5.10. Then $\mathcal{F}$ is a stack according to Definition 5.12 if and only if F is a stack according to Definition 1.1, that is, morphisms and objects glue if and only if for all $U \in \mathrm{Open}(X)$ and for all open covers $\mathcal{U} = \{U_i\}$ of U, the functor

$$F(U) \longrightarrow \text{2-lim}\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

given by the universal property of the 2-limit in Grpd is an equivalence of categories.

Proof. The main idea behind the proof is that (1) objects glue if and only if the functor

$$F(U) \longrightarrow \text{2-lim}\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

is essentially surjective, and (2) morphisms glue if and only if it is fully faithful.

Before proving these two conditions, let's recall the category of descent data from Definition 4.9. Recall that

$$\mathrm{Desc}(\mathcal{U}, F) = \text{2-lim}\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k} F(U_i \cap U_j \cap U_k) \right)$$

is the groupoid with:

- ▷ objects (a_i, ϕ_{ij}), where $a_i \in F(U_i)$ and $\phi_{ij} : a_i \mid_{U_{ij}} \xrightarrow{\cong} a_j \mid_{U_{ij}}$ in $F(U_{ij})$, such that $\phi_{ij}^{-1} = \phi_{ji}$ and the cocycle condition $\phi_{ij} \mid_{U_{ijk}} \circ \phi_{jk} \mid_{U_{ijk}} = \phi_{ik} \mid_{U_{ijk}}$ in $F(U_{ijk})$ is satisfied,
- ▷ isomorphisms $f : (a_i, \phi_{ij}) \xrightarrow{\cong} (a'_i, \phi'_{ij})$, which consist of isomorphisms $f_i : a_i \xrightarrow{\cong} a'_i$ in $F(U_i)$ such that the diagrams

$$\begin{array}{ccc} a_i \mid_{U_{ij}} & \xrightarrow{f_i|_{U_{ij}}} & a'_i \mid_{U_{ij}} \\ \downarrow{\scriptstyle \phi_{ij}} & & \downarrow{\scriptstyle \phi'_{ij}} \\ a_j \mid_{U_{ij}} & \xrightarrow[f_j|_{U_{ij}}]{} & a'_j \mid_{U_{ij}} \end{array}$$

commute in $F(U_{ij})$.

Next, we should say what the functor $F(U) \to \mathrm{Desc}(\mathcal{U}, F)$ is. We will call it the comparison functor $R : F(U) \to \mathrm{Desc}(\mathcal{U}, F)$. Let's start by defining how it maps objects. For $a \in F(U)$, we have $R(a) = (a \mid_{U_i}, \theta_{ij}(a)) \in \mathrm{Desc}(\mathcal{U}, F)$, where $a \mid_{U_i} \in F(U_i)$ and $\theta_{ij}(a) : (a \mid_{U_i}) \mid_{U_{ij}} \xrightarrow{\cong} (a \mid_{U_j}) \mid_{U_{ij}}$ are canonical isomorphisms given by the pseudofunctoriality of F. To more precise, for the composable pair of inclusions $U_{ij} \hookrightarrow U_i \hookrightarrow U$, since F is a contravariant pseudofunctor we have a specified natural isomorphism $c_i^{ij} : F(U_{ij} \hookrightarrow U_i \hookrightarrow U) \cong F(U_{ij} \hookrightarrow U_i) \circ F(U_i \hookrightarrow U)$. We can picture this in:

$$\begin{array}{ccc} F(U) & \xrightarrow{F(U_i \hookrightarrow U)} & F(U_i) \\ & \overset{c_i^{ij}}{\swarrow} & \downarrow{\scriptstyle F(U_{ij} \hookrightarrow U_i)} \\ & {\scriptstyle F(U_{ij} \hookrightarrow U)} \searrow & \\ & & F(U_{ij}). \end{array}$$

Similarly, for the composable pair of inclusions $U_{ij} \hookrightarrow U_j \hookrightarrow U$, we have a specified natural isomorphism $c_j^{ij} : F(U_{ij} \hookrightarrow U_j \hookrightarrow U) \cong F(U_{ij} \hookrightarrow U_j) \circ F(U_j \hookrightarrow U)$. We can picture this in

$$\begin{array}{ccc} & F(U) & \\ & \swarrow \quad \overset{c_j^{ij}}{\swarrow} & \searrow^{F(U_{ij} \hookrightarrow U)} \\ {\scriptstyle F(U_i \hookrightarrow U)}\downarrow & & \\ F(U_j) & \xrightarrow[F(U_{ij} \hookrightarrow U_j)]{} & F(U_{ij}). \end{array}$$

Then, c_i^{ij} has an inverse $(c_i^{ij})^{-1} : (U_{ij} \hookrightarrow U_i) \circ F(U_i \hookrightarrow U) \cong F(U_{ij} \hookrightarrow U_i \hookrightarrow U)$ which we can picture in:

$$\begin{array}{ccc} F(U) & \xrightarrow{F(U_i \hookrightarrow U)} & F(U_i) \\ & \underset{(c_i^{ij})^{-1}}{\swarrow} & \downarrow{\scriptstyle F(U_{ij} \hookrightarrow U_i)} \\ & {\scriptstyle F(U_{ij} \hookrightarrow U)} \searrow & \\ & & F(U_{ij}). \end{array}$$

Since $U_{ij} \hookrightarrow U_i \hookrightarrow U = U_{ij} \hookrightarrow U_j \hookrightarrow U = U_{ij} \hookrightarrow U$, so $F(U_{ij} \hookrightarrow U_i \hookrightarrow U) = F(U_{ij} \hookrightarrow U_j \hookrightarrow U)$, we can horizontally compose $(c_i^{ij})^{-1}$ with c_j^{ij} to define a natural isomorphism $\theta_{ij} = c_j^{ij} \circ (c_i^{ij})^{-1} : F(U_{ij} \hookrightarrow U_i) \circ F(U_i \hookrightarrow U) \cong F(U_{ij} \hookrightarrow U_j) \circ F(U_j \hookrightarrow U)$ which we can picture in:

$$\begin{array}{ccc} F(U) & \xrightarrow{F(U_i \hookrightarrow U)} & F(U_i) \\ {\scriptstyle F(U_j \hookrightarrow U)}\downarrow & \swarrow\!\!\!\swarrow \; \theta_{ij} \; \nearrow\!\!\!\nearrow & \downarrow{\scriptstyle F(U_{ij} \hookrightarrow U_i)} \\ F(U_j) & \xrightarrow[F(U_{ij} \hookrightarrow U_j)]{} & F(U_{ij}). \end{array}$$

The component of θ_{ij} at $a \in F(U)$ is the canonical isomorphism $\theta_{ij}(a) : (a \mid_{U_i}) \mid_{U_{ij}} \xrightarrow{\cong} (a \mid_{U_j}) \mid_{U_{ij}}$.

We need to check that the inverses satisfy $(\theta_{ij}(a))^{-1} = \theta_{ji}(a)$. We have $\theta_{ij}^{-1} = (c_j^{ij} \circ (c_i^{ij})^{-1})^{-1} = c_i^{ij} \circ (c_j^{ij})^{-1}$, which is how we define θ_{ji} by swapping the roles of i and j in our previous reasoning, so $\theta_{ij}^{-1} = \theta_{ji}$, and the component at $a \in F(U)$ satisfies $\theta_{ij}^{-1}(a) = (\theta_{ij}(a))^{-1} = \theta_{ji}(a)$.

We need to check that the $\theta_{ij}(a)$ satisfy the cocycle condition on triple overlaps, i.e. that

$$\theta_{ij}(a) \mid_{U_{ijk}} \circ \theta_{jk}(a) \mid_{U_{ijk}} = \theta_{ik}(a) \mid_{U_{ijk}} .$$

I'm not sure how to show this.

Next, let's define how R maps isomorphisms. For $f : a \xrightarrow{\cong} b$ in $F(U)$, we have $R(f) : R(a) = (a \mid_{U_i}, \theta_{ij}(a)) \xrightarrow{\cong} R(b) = (b \mid_{U_i}, \theta_{ij}(b))$, which consists of $R(f)_i = f \mid_{U_i} : a \mid_{U_i} \xrightarrow{\cong} b \mid_{U_i}$ in $F(U_i)$. Note that the squares

$$\begin{array}{ccc} (a \mid_{U_i}) \mid_{U_{ij}} & \xrightarrow{(f|_{U_i})|_{U_{ij}}} & (b \mid_{U_i}) \mid_{U_{ij}} \\ {\scriptstyle \theta_{ij}(a)}\downarrow & & \downarrow{\scriptstyle \theta_{ij}(b)} \\ (a \mid_{U_j}) \mid_{U_{ij}} & \xrightarrow[(f|_{U_j})|_{U_{ij}}]{} & (b \mid_{U_j}) \mid_{U_{ij}} \end{array}$$

in $F(U_{ij})$ commute since $\theta_{ij} : F(U_{ij} \hookrightarrow U_i) \circ F(U_i \hookrightarrow U) \cong F(U_{ij} \hookrightarrow U_j) \circ F(U_j \hookrightarrow U)$ is a natural isomorphism of functors from $F(U) \to F(U_{ij})$.

Next, we should say what the prestack corresponding to the pseudofunctor F is. Recall from Theorem 5.10 that the correspondence between pseudofunctors from $\mathrm{Open}(X) \to \mathrm{Cat}$ and prestacks over $\mathrm{Open}(X)$ is given by the Grothendieck construction. Therefore, the prestack corresponding to F is given by the 1-category $\int F$, which is the data of:

- ▷ objects (U, a), where $U \in \mathrm{Open}(X)$ and $a \in F(U)$,
- ▷ morphisms $(V, a) \to (U, b)$ which consist of the inclusion $V \hookrightarrow U$ in $\mathrm{Open}(X)$ and a morphism $a \to F(V \hookrightarrow U)(b) = b \mid_V$ in $F(V)$,

and the projection functor $\pi : \int F \to \mathrm{Open}(X)$ sending $(U, a) \mapsto U$ and $((V, a) \to (U, b)) \mapsto V \hookrightarrow U$.

We are finally ready to prove the equivalence between the two conditions:

(1) ($\Rightarrow$) Suppose objects glue. We want to show that R is essentially surjective. Let $(a_i, \psi_{ij}) \in \mathrm{Desc}(\mathcal{U}, F)$, where $a_i \in F(U_i)$, and $\psi_{ij} : a_i \mid_{U_{ij}} \xrightarrow{\cong} a_j \mid_{U_{ij}}$ in $F(U_{ij})$ satisfying the cocycle condition on triple overlaps. Therefore, we have $(U_i, a_i) \in \int F$ with $(U_i, a_i) \in \pi^{-1}(U_i)$, and isomorphisms $\psi_{ij}^+ : (U_i, a_i) \mid_{U_{ij}} = (U_{ij}, a_i \mid_{U_{ij}}) \xrightarrow{\cong} (U_j, a_j) \mid_{U_{ij}} = (U_{ij}, a_j \mid_{U_{ij}})$ in $\int F$ consisting of the identity maps $\mathrm{id}_{U_{ij}}$ and the isomorphisms ψ_{ij}. We know that the ψ_{ij}^+ satisfy the cocycle condition on triple overlaps since the ψ_{ij} do. Then, since objects glue, there exists an object $(U, a) \in \pi^{-1}(U)$ for some $a \in F(U)$, and isomorphisms $\phi_i^+ : (U, a) \mid_{U_i} = (U_i, a \mid_{U_i}) \xrightarrow{\cong} (U_i, a_i)$

with $\pi(\phi_i^+) = \mathrm{id}_{U_i}$ such that $\phi_j^+ \mid_{U_{ij}} = \psi_{ij}^+ \circ \phi_i^+ \mid_{U_{ij}}$. Now, consider $R(a) = (a \mid_{U_i}, \theta_{ij}(a)) \in \mathrm{Desc}(\mathcal{U}, F)$. We know that in the data of ϕ_i^+ we have isomorphisms $\phi_i : a \mid_{U_i} \xrightarrow{\cong} a_i$, and we claim that they give rise to an isomorphism $R(a) = (a \mid_{U_i}, \theta_{ij}(a)) \xrightarrow{\cong} (a_i, \psi_{ij})$ in $\mathrm{Desc}(\mathcal{U}, F)$. It suffices to show that the following squares in $F(U_{ij})$ commute:

$$\begin{array}{ccc} (a \mid_{U_i}) \mid_{U_{ij}} & \xrightarrow{\phi_i|_{U_{ij}}} & a_i \mid_{U_{ij}} \\ {\scriptstyle \theta_{ij}(a)}\downarrow & & \downarrow{\scriptstyle \psi_{ij}} \\ (a \mid_{U_j}) \mid_{U_{ij}} & \xrightarrow[\phi_j|_{U_{ij}}]{} & a_j \mid_{U_{ij}}. \end{array}$$

Since $\phi_j^+ \mid_{U_{ij}} = \psi_{ij}^+ \circ \phi_i^+ \mid_{U_{ij}}$ in $\int F$, it follows that $\phi_j \mid_{U_{ij}} = \psi_{ij} \circ \phi_i \mid_{U_{ij}}$ holds in $F(U_{ij})$ up to composing with a canonical isomorphism given by the pseudofunctoriality of F. Since $\theta_{ij}(a)$ is such an isomorphism, we have $\phi_j \mid_{U_{ij}} \circ \theta_{ij}(a) = \psi_{ij} \circ \phi_i \mid_{U_{ij}}$, and so the squares commute.

($\Leftarrow$) Suppose R is essentially surjective. We want to show that objects glue. Let $(U_i, a_i) \in \pi^{-1}(U_i)$ for some $a_i \in F(U_i)$, and let $\psi_{ij}^+ : (U_i, a_i) \mid_{U_{ij}} = (U_{ij}, a_i \mid_{U_{ij}}) \xrightarrow{\cong} (U_j, a_j) \mid_{U_{ij}} = (U_{ij}, a_j \mid_{U_{ij}})$ be isomorphisms satisfying the cocycle condition on triple overlaps. We know that in the data of the ψ_{ij}^+ we have isomorphisms $\psi_{ij} : a_i \mid_{U_{ij}} \xrightarrow{\cong} a_j \mid_{U_{ij}}$ in $F(U_{ij})$, which satisfy the cocycle condition on triple overlaps since the ψ_{ij}^+ do. Therefore, we have an object $(a_i, \psi_{ij}) \in \mathrm{Desc}(\mathcal{U}, F)$. Since R is essentially surjective, there exists $a \in F(U)$ such that $R(a) = (a \mid_{U_i}, \theta_{ij}(a)) \cong (a_i, \psi_{ij})$. In the data of this isomorphism, we have isomorphisms $\phi_i : a \mid_{U_i} \xrightarrow{\cong} a_i$ such that the following squares commute:

$$\begin{array}{ccc} (a \mid_{U_i}) \mid_{U_{ij}} & \xrightarrow{\phi_i|_{U_{ij}}} & a_i \mid_{U_{ij}} \\ {\scriptstyle \theta_{ij}(a)}\downarrow & & \downarrow{\scriptstyle \psi_{ij}} \\ (a \mid_{U_j}) \mid_{U_{ij}} & \xrightarrow[\phi_j|_{U_{ij}}]{} & a_j \mid_{U_{ij}}. \end{array}$$

Since $a \in F(U)$ we have $a \in \pi^{-1}(U)$. Then, the data of the isomorphisms ϕ_i in $F(U_i)$ along with identity maps id_{U_i} give rise to isomorphisms $\phi_i^+ : (U_i, a \mid_{U_i}) \xrightarrow{\cong} (U_i, a_i)$ in $\int F$, with $\pi(\phi_i^+) = \mathrm{id}_{U_i}$. All that remains is to show that $\phi_j^+ \mid_{U_{ij}} = \psi_{ij}^+ \circ \phi_i^+ \mid_{U_{ij}}$. Since the above square commutes, we have $\phi_j \mid_{U_{ij}} \circ \theta_{ij}(a) = \psi_{ij} \circ \phi_i \mid_{U_{ij}}$, so we have $\phi_j \mid_{U_{ij}} = \psi_{ij} \circ \phi_i \mid_{U_{ij}}$ up to composing with the canonical isomorphism $\theta_{ij}(a)$, and so it follows that $\phi_j^+ \mid_{U_{ij}} = \psi_{ij}^+ \circ \phi_i^+ \mid_{U_{ij}}$.

(2) ($\Rightarrow$) Suppose morphisms glue. We want to show that R is fully faithful. Let $a, b \in F(U)$, and consider the map $\mathrm{Hom}_{F(u)}(a, b) \to \mathrm{Hom}_{\mathrm{Desc}(\mathcal{U},F)}(R(a), R(b))$ sending $f \mapsto R(f)$. We want to show that this map is a bijection, i.e. that for every isomorphism $\psi : R(a) \xrightarrow{\cong} R(b)$ in $\mathrm{Hom}_{\mathrm{Desc}(\mathcal{U},F)}(R(a), R(b))$, there exists a unique isomorphism $\phi \in \mathrm{Hom}_{F(u)}(a, b)$ such that $R(\phi) = \psi$. Let $\psi : R(a) = (a \mid_{U_i}, \theta_{ij}(a)) \xrightarrow{\cong} R(b) = (b \mid_{U_i}, \theta_{ij}(b))$ be an arbitrary isomorphism, consisting of isomorphisms $\psi_i : a \mid_{U_i} \xrightarrow{\cong} b \mid_{U_i}$ in $F(U_i)$ such that the following

squares commute:

$$\begin{array}{ccc} (a\mid_{U_i})\mid_{U_{ij}} & \xrightarrow{\psi_i|_{U_{ij}}} & (b\mid_{U_i})\mid_{U_{ij}} \\ \Big\downarrow{\scriptstyle \theta_{ij}(a)} & & \Big\downarrow{\scriptstyle \theta_{ij}(b)} \\ (a\mid_{U_j})\mid_{U_{ij}} & \xrightarrow[\psi_j|_{U_{ij}}]{} & (b\mid_{U_j})\mid_{U_{ij}}. \end{array}$$

Therefore, $\psi_i\mid_{U_{ij}} = \psi_j\mid_{U_{ij}}$ up to composing with the canonical isomorphisms $\theta_{ij}(a), \theta_{ij}(b)$. Consider objects $(U,a),(U,b) \in \int F$ with $\pi(U,a) = \pi(U,b) = U$. The data of the ψ_i together with the identity maps id_{U_i} give rise to morphisms $\psi_i^+ : (U,a)\mid_{U_i} = (U_i, a\mid_{U_i}) \to (U,b)\mid_{U_i} = (U_i, b\mid_{U_i})$ in $\int F$. Since $\psi_i\mid_{U_{ij}} = \psi_j\mid_{U_{ij}}$ up to composing with canonical isomorphisms, we have $\psi_i^+\mid_{U_{ij}} = \psi_j^+\mid_{U_{ij}}$. Then, since morphisms glue, there exists a unique morphism $\phi^+ : (U,a) \to (U,b)$ such that $\pi(\phi^+) = \mathrm{id}_U$ and $\phi^+\mid_{U_i} = \psi_i^+$. In the data of the morphism $\phi^+ : (U,a) \to (U,b)$, we have the data of an isomorphism $\phi : a \xrightarrow{\cong} b$ in $F(U)$, and so the isomorphism $R(\phi) : R(a) \xrightarrow{\cong} R(b)$ consists of $\phi\mid_{U_i} : a\mid_{U_i} \xrightarrow{\cong} b\mid_{U_i}$. Since $\phi^+\mid_{U_i} = \psi_i^+$, we have $\phi\mid_{U_i} = \psi_i$, so we can see that $R(\phi) = \psi$, and so we have shown existence. It remains to show uniqueness. Suppose there exists $\eta : a \xrightarrow{\cong} b$ in $F(U)$ such that $R(\eta) = \psi$. Then, we have $\eta\mid_{U_i} = \psi_i$. The data of η and id_U gives rise to a morphism $\eta^+ : (U,a) \to (U,b)$ in $\int F$ such that $\pi(\eta^+) = \mathrm{id}_U$ and $\eta^+\mid_{U_i} = \psi_i^+$, but we already said such a morphism is unique since morphisms glue, so $\eta^+ = \phi^+$, which implies $\eta = \phi$, which shows uniqueness.

($\Leftarrow$) Suppose R is fully faithful. We want to show that morphisms glue. Let $(U,a),(U,b) \in \int F$ for some $a,b \in F(U)$, so $\pi(U,a) = \pi(U,b) = U$. Let $\phi_i^+ : (U,a)\mid_{U_i} = (U_i, a\mid_{U_i}) \to (U,b)\mid_{U_i} = (U_i, b\mid_{U_i})$ be morphisms such that $\phi_i^+\mid_{U_{ij}} = \phi_j^+\mid_{U_{ij}}$. We want to show there exists a unique morphism $\phi^+ : (U,a) \to (U,b)$ so that $\pi(\phi^+) = \mathrm{id}_U$ and $\phi^+\mid_{U_i} = \phi_i^+$. We know that in the data of the ϕ_i^+ we have isomorphisms $\phi_i : a\mid_{U_i} \xrightarrow{\cong} b\mid_{U_i}$. Since $\phi_i^+\mid_{U_{ij}} = \phi_j^+\mid_{U_{ij}}$, we have that $\phi_i\mid_{U_{ij}} = \phi_j\mid_{U_{ij}}$ up to composing with the canonical isomorphisms $\theta_{ij}(a), \theta_{ij}(b)$, so the following squares commute:

$$\begin{array}{ccc} (a\mid_{U_i})\mid_{U_{ij}} & \xrightarrow{\psi_i|_{U_{ij}}} & (b\mid_{U_i})\mid_{U_{ij}} \\ \Big\downarrow{\scriptstyle \theta_{ij}(a)} & & \Big\downarrow{\scriptstyle \theta_{ij}(b)} \\ (a\mid_{U_j})\mid_{U_{ij}} & \xrightarrow[\psi_j|_{U_{ij}}]{} & (b\mid_{U_j})\mid_{U_{ij}}. \end{array}$$

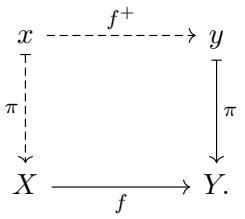
Therefore, the ϕ_i form the data of an isomorphism $\phi : R(a) = (a\mid_{U_i}, \theta_{ij}(a)) \xrightarrow{\cong} R(b) = (b\mid_{U_i}, \theta_{ij}(b))$. Since R is fully faithful, the map $\mathrm{Hom}_{F(u)}(a,b) \to \mathrm{Hom}_{\mathrm{Desc}(\mathcal{U},F)}(R(a),R(b))$ defined by $f \mapsto R(f)$ is a bijection, so there exists a unique $\psi : a \xrightarrow{\cong} b$ in $F(U)$ such that $R(\psi) = \phi$. Therefore, we have $\psi\mid_{U_i} = \phi_i$. The data of ψ and id_U give rise to a morphism $\psi^+ : (U,a) \to (U,b)$ such that $\pi(\psi^+) = \mathrm{id}_U$. Then, since $\psi\mid_{U_i} = \phi_i$, it follows that $\psi^+\mid_{U_i} = \phi_i^+$, so we have shown existence. It remains to show uniqueness. Suppose we have $\eta^+ : (U,a) \to (U,b)$ so $\pi(\eta^+) = \mathrm{id}_U$ and $\eta^+\mid_{U_i} = \phi_i^+$. In the data of η we have an isomorphism $\eta : a \xrightarrow{\cong} b$, so we can consider $R(\eta) : R(a) \xrightarrow{\cong} R(b)$ which consists of $\eta\mid_{U_i}$. Then, since $\eta^+\mid_{U_i} = \phi_i^+$ it follows that $\eta\mid_{U_i} = \phi_i$, so $R(\eta) = \phi$. Since R is fully faithful, this implies that $\eta = \psi$, which in turn implies that $\eta^+ = \psi^+$, which shows uniqueness.

□

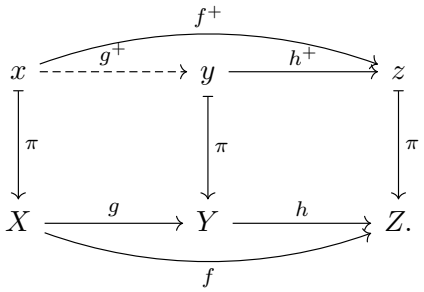
Remark 5.16. For the sake of completeness, we restate the important definitions and theorems from this section for the global classical topology.

Definition 5.17 (Prestack)**.** A *prestack* over Top is a 1-category $\mathcal{F}$ and a functor $\pi : \mathcal{F} \to \text{Top}$ satisfying the following two conditions:

(1) Pullbacks exist: if $f : X \to Y$ is a continuous map and $y \in \pi^{-1}(Y)$, then there exists $x \in \pi^{-1}(X)$ and $f^+ : x \to y$ so that $\pi(f^+) = f$. We say $f^+ : x \to y$ is a morphism over f, and we have



(2) Universal property for pullbacks: suppose we have composable continuous maps $X \xrightarrow{g} Y \xrightarrow{h} Z$ with $hg = f : X \to Z$ and $z \in \pi^{-1}(Z)$, then we know by (1) that there exists $x \in \pi^{-1}(X)$, $y \in \pi^{-1}(Y)$, $f^+ : x \to z$, and $h^+ : y \to z$ so that $\pi(f^+) = f$ and $\pi(h^+) = h$. The universal property for pullbacks says that there exists a unique $g^+ : x \to y$ such that $\pi(g^+) = g : X \to Y$ and that the following diagram commutes:

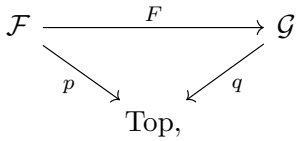


This definition is adapted from Definition 1 in Alper's *Evolution of Stacks and Moduli* [Alp].

Remark 5.18. Prestacks over Top are exactly the same as fibrations over Top.

Definition 5.19 (Category of fibrations)**.** There is a *category of fibrations* (or equivalently prestacks) over Top, which we will denote by $\text{Fib}(\text{Top}) = \text{Prestack}(\text{Top})$, which is the data of:

- ▷ objects which are fibrations (prestacks) $p : \mathcal{F} \to \text{Top}$, $q : \mathcal{G} \to \text{Top}$, $\cdots$, and
- ▷ morphisms $p \to q$ which are functors $F : \mathcal{F} \to \mathcal{G}$ that commute with the projections, that is, such that the following diagrams commute:



and which send Cartesian morphisms in $\mathcal{F}$ to Cartesian morphisms in $\mathcal{G}$. These are called *Cartesian functors*. This definition is adapted from Definition 9.1.14 and Theorem 9.1.18 in [JY21].

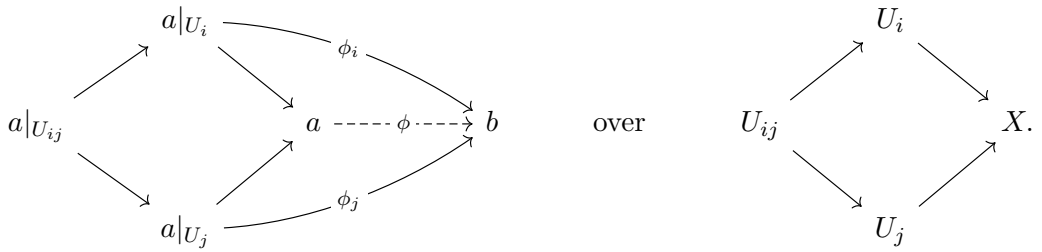
Theorem 5.20. There is an equivalence of categories

$$\text{Fun}^{\text{ps}}(\text{Top}^{\text{op}}, \text{Cat}) \simeq \text{Prestack}(\text{Top})$$

given by the Grothendieck construction.

Definition 5.21 (Stack, prestack version)**.** We say a prestack $\pi : \mathcal{F} \to \text{Top}$ is a *stack* if:

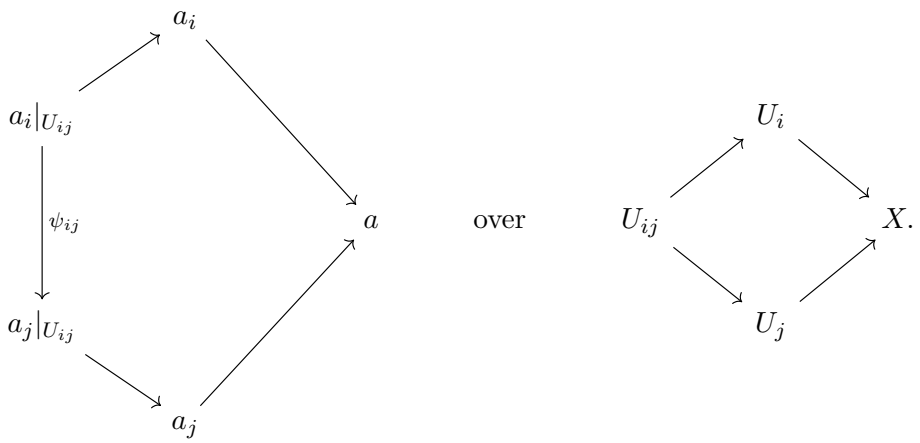
(1) Morphisms glue: for $a, b \in \mathcal{F}$ so that $\pi(a) = \pi(b) = X$, a jointly surjective open embedding $\mathcal{U} = \{U_i \hookrightarrow X\}$, and morphisms $\phi_i : a|_{U_i} \to b\,|_{U_i}$ such that $\phi_i|_{U_{ij}} = \phi_j|_{U_{ij}}$, there exists a unique morphism $\phi : a \to b$ so that $\pi(\phi) = \mathrm{id}_X$ and $\phi\,|_{U_i} = \phi_i$. We can picture this as



(2) Objects glue: for $a_i \in \pi^{-1}(U_i)$ and isomorphisms $\psi_{ij} : a_i|_{U_{ij}} \xrightarrow{\cong} a_j|_{U_{ij}}$ satisfying the cocycle condition on triple overlaps

$$\psi_{jk}|_{U_{ijk}} \circ \psi_{ij}|_{U_{ijk}} = \psi_{ik}|_{U_{ijk}},$$

there exists an object $a \in \pi^{-1}(X)$ and isomorphisms $\phi_i : a|_{U_i} \xrightarrow{\cong} a_i$ so that $\pi(\phi_i) = \mathrm{id}_{U_i}$ and $\phi_j|_{U_{ij}} = \psi_{ij} \circ \phi_i|_{U_{ij}}$. We can picture this as



This definition is adapted from Definition 3 in Alper's *Evolution of Stacks and Moduli* [Alp].

Theorem 5.22. Let $F : \mathrm{Top}^{\mathrm{op}} \to \mathrm{Grpd}$ be a pseudofunctor corresponding to a prestack $\pi : \mathcal{F} \to \mathrm{Top}$ under the equivalence of categories in Theorem 5.20. Then $\mathcal{F}$ is a stack according to Definition 5.21 if and only if F is a stack according to Definition 2.6, that is, morphisms and objects glue if and only if for every space X and for every set of jointly surjective open embeddings $\mathcal{U} = \{U_i \hookrightarrow X\}$, the functor

$$F(X) \longrightarrow \text{2-lim}\left(\prod F(U_i) \rightrightarrows \prod F(U_i \cap U_j) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod F(U_i \cap U_j \cap U_k) \right)$$

given by the universal property of the 2-limit in Grpd is an equivalence of categories.

6. Example: Principal G-Bundles

Now that we have two definitions of a stack to work with, we are ready to give a concrete example and prove that it's a stack using both definitions.

Example 6.1. Let $F : \mathrm{Top}^{\mathrm{op}} \to \mathrm{Grpd}$ be the assignment of:

- ▷ every space $X \in \mathrm{Top}$ to its groupoid of principal G-bundles $F(X) = \mathrm{Prin}_G(X)$, and
- ▷ every continuous map $f : X \to Y$ the functor $F(f) : \mathrm{Prin}_G(Y) \to \mathrm{Prin}_G(X)$ defined:

(1) on objects by sending $P \in \mathrm{Prin}_G(Y)$ to the pullback $F(f)(P) = X \times_Y P \in \mathrm{Prin}_G(X)$ of P along f, seen as

$$\begin{array}{ccc} X \times_Y P & \longrightarrow & P \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \pi} \\ X & \xrightarrow[f]{} & Y, \end{array}$$

(2) on isomorphisms by sending $\varphi : P \xrightarrow{\cong} P'$ in $\mathrm{Prin}_G(Y)$ to the isomorphism $F(f)(\varphi) : X \times_Y P \xrightarrow{\cong} X \times_Y P'$ in $\mathrm{Prin}_G(X)$ defined by

$$(x, p) \mapsto (x, \varphi(p)) = F(f)(\varphi)(x, p).$$

Note this is well-defined since $\pi'(\varphi(p)) = \pi(p) = f(x)$.

Proposition 6.2. The assignment $F : \mathrm{Top}^{\mathrm{op}} \to \mathrm{Grpd}$ is a pseudofunctor.

Proof. To show this by definition, according to Definition 3.7, we would have to further specify a bunch of natural isomorphisms and show they satisfy the two coherence conditions, which is rather tedious. Instead, we will show that there is a prestack corresponding to F under the equivalence of categories in Theorem 5.17, which is given by the Grothendieck construction. □

Remark 6.3. We can now prove that the pseudofunctor F and the prestack $\pi : \int F \to \mathrm{Top}$ are both stacks.

Proposition 6.4. The prestack $\pi : \int F \to \mathrm{Top}$ is a stack.

Proof. □

Proposition 6.5. The pseudofunctor F is a stack.

Proof. □

References

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